

**DEPARTMENT OF TRANSPORTATION
OFFICE OF THE ASSISTANT SECRETARY FOR RESEARCH AND TECHNOLOGY
VOLPE NATIONAL TRANSPORTATION SYSTEMS CENTER (VOLPE CENTER)
TECHNOLOGY, INNOVATION AND STRATEGY DIVISION (V336)**

SOLE SOURCE JUSTIFICATION

- A. Description of Supplies/Services.** The Volpe National Transportation Systems Center (Volpe Center), in collaboration with the Intelligent Transportation Systems Joint Program Office (ITS JPO) and the Office of the Assistant Secretary for Research and Technology (OST-R), is investigating methods for the federal government to partner with private industry to leverage disparate data streams to create a “real-time” picture of freight carrier legitimacy. Current FMCSA datasets – Motor Carrier Management Information System (MCMIS), Safety Measurement System (SMS) -- are highly effective for rated carriers but offer limited visibility into the thousands of new and unrated entrants that dominate the spot market. While the vast majority of motor carriers without a safety rating are American trucking companies (especially small owner operators and new fleets), there are also gaps among Mexican-domiciled carriers committing cabotage. To support the investigation of this problem, this project seeks to fuse public FMCSA data with private industry data to create a predictive actuarial model that acts as a surrogate for traditional safety ratings.

The objective of this project is to develop and validate a predictive actuarial model that assigns an Interim Risk Rating to unrated Mexican-domiciled carriers with authorities to operate along the US-Mexico commercial border zone or to conduct long-haul international transport. This rating will serve as a statistically valid proxy for FMCSA safety ratings, segmenting Mexican-domiciled carriers with these authorities into distinct risk pools based on their probability of safety incidents and FMCSA violations.

- B. Market Research.** Market research was conducted to identify companies capable of developing and validating a predictive actuarial model that assigns an Interim Risk Rating to unrated Mexican-domiciled carriers. The solution must utilize proprietary AI to aggregate non-traditional leading indicators of risk – such as crash scores, and “chameleon” carrier identifiers – to serve as a statistically valid proxy for FMCSA safety ratings. Furthermore, the solution must be able to segment carriers operating along the U.S.-Mexico commercial border zone and long-haul corridors into distinct risk pools based on their probability of cabotage and safety violations.

While Bluewire’s model is built to bridge information gaps, its main competitors, such as Central Analysis Bureau (CAB) and SambaSafety, frequently fail to meet this specific requirement because they are fundamentally reactive rather than predictive. These platforms primarily function as data aggregators that rely on trailing indicators, meaning they require a substantial volume of existing FMCSA inspection and violation history to generate a meaningful score. For unrated Mexican-domiciled carriers or new entrants with limited U.S. road time, these systems often return an “Incomplete” or “Not Rated” status. This creates a data gap that makes it difficult to assign a statistically valid Interim Risk Rating, effectively leaving these carriers invisible to traditional risk assessment tools until a major incident or audit occurs.

Furthermore, many traditional competitors lack the sophisticated leading-indicator logic required to identify freight behavior in the absence of DOT data. While Bluewire utilizes proprietary AI to conduct text mining on social sentiment, recruitment behaviors, and “chameleon carrier” identifiers, competitors typically stick to rigid, structured datasets like MCMIS or MVR records. Without these non-traditional

data points, other models cannot detect the latent risk of a carrier that may have irregular behavior or poor safety management but hasn't yet been inspected. This makes these other competitors insufficient for the objective of segmenting Mexican-domiciled carriers into risk pools, as they cannot differentiate between a “legitimate but unrated” carrier and a “bad acting but unrated” one.

Bluewire is a small business that is not part of a GSA, NASA SEWP, or FAA SAVES contract. The company is only accessible via direct award.

C. Supporting Rationale.

(1) Sole Source: Only one responsible source and no other supplies or services will satisfy agency requirements.

Bluewire satisfies the project requirements by transforming the traditional, often binary, safety rating system into a multidimensional GAP Score that proactively identifies “severity risk” rather than just historical compliance. For unrated Mexican-domiciled carriers, this is particularly valuable because Bluewire’s model does not rely on a completed FMCSA audit to generate a score. Instead, it aggregates data across nine critical vulnerability gaps, such as MCS-150 filing accuracy, driver out-of-service (OOS) rates, and legal exposure, to create a 0–100 weighted rating. This allows them to assign an Interim Risk Rating to carriers that would otherwise remain invisible in traditional safety databases, providing an immediate basis for risk pooling based on their likelihood of being involved in high-severity incidents.

The statistical validity of Bluewire's model is anchored in its use of proprietary AI and dynamic peer-grouping. Unlike the FMCSA’s static Safety Measurement System (SMS), which can be slow to update and often penalizes smaller fleets unfairly, Bluewire’s AI analyzes more than 750,000 DOT numbers to create trend lines based on recent performance data. By comparing Mexican-domiciled carriers against similar-sized peers operating in the same cross-border corridors, the model accounts for the specific operational realities of international long-haul transport. This satisfies the objective for a statistically valid proxy, as it uses weighted variables—including crash recency, violation patterns in high-risk categories (like Hours of Service or Driver Fitness), and even sentiment analysis—to predict future safety or cabotage outcomes with high precision.

Ultimately, Bluewire’s approach enables the precise segmentation of carriers into distinct risk pools by identifying latent vulnerabilities that traditional FMCSA ratings miss. For carriers operating in the US-Mexico border zone, the model can flag “chameleon” behavior or deteriorating safety cultures before a formal intervention occurs. Bluewire’s approach effectively bridges the information gap for unrated carriers, allowing them to validate an interim rating that aligns with the same safety standards required for U.S.-based domestic fleets.

(2) Brand Name: N/A

(3) Unusual and Compelling Urgency: N/A

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